

JAVA STRING HANDLING AND COLLECTION FRAMEWORK

Assessment Tool - 2

Name : Tamil Elakiya S

Reg No : 192524443

1 . Design an iterator for a custom collection class. Specify its state, hasNext(), next(), removal policy and expected behaviour at the end of iteration.

Aim :

Design an iterator for a custom collection class that:

- Maintains the current position (state) in the collection.
- Uses hasNext() to check if more elements are available.
- Uses next() to return the next element.
- Does **not allow removal** of elements (read-only iterator).
- Throws an exception when next() is called after the last element.

Concepts Used

- Iterator Interface
- ArrayList
- hasNext()
- next()
- UnsupportedOperationException
- NoSuchElementException

Method Signature

- public boolean hasNext()
- public String next()
- public void remove()

Code :

```
import java.util.ArrayList;

import java.util.Iterator;

import java.util.NoSuchElementException;

class CustomCollection {

    private ArrayList<String> items = new ArrayList<>();

    public void add(String item) {
        items.add(item);
    }

    public Iterator<String> iterator() {
        return new CustomIterator();
    }

    private class CustomIterator implements Iterator<String> {

        private int currentIndex = 0;

        public boolean hasNext() {
            return currentIndex < items.size();
        }

        public String next() {
            if (!hasNext()) {
                throw new NoSuchElementException("End of Collection Reached");
            }
        }
    }
}
```

```
        return items.get(currentIndex++);
    }

    public void remove() {
        throw new UnsupportedOperationException("Removal is Not Allowed");
    }
}

}
```

```
public class IteratorDemo {

    public static void main(String[] args) {
        CustomCollection collection = new CustomCollection();
        collection.add("Apple");
        collection.add("Banana");
        collection.add("Mango");

        Iterator<String> itr = collection.iterator();
        System.out.println("Collection Elements:");
        while (itr.hasNext()) {
            System.out.println(itr.next());
        }

        try {
            itr.next();
        } catch (NoSuchElementException e) {
            System.out.println("\nEnd of Collection Reached!");
        }
    }
}
```

```
try {  
    itr.remove();  
} catch (UnsupportedOperationException e) {  
    System.out.println("Removal Not Allowed!");  
}  
}  
}
```

Output :

```
D:\JavaPrograms>javac IteratorDemo.java  
  
D:\JavaPrograms>java IteratorDemo  
Collection Elements:  
Apple  
Banana  
Mango  
  
End of Collection Reached!  
Removal Not Allowed!
```

Explanation :

- The iterator maintains its state using currentIndex.
- hasNext() checks whether another element exists in the collection.
- next() returns the current element and moves the iterator to the next position.
- If next() is called after the last element, it throws a NoSuchElementException.
- The remove() method throws UnsupportedOperationException, making the iterator read-only.

Purpose

A custom iterator is used to access collection elements one by one without exposing the internal structure of the collection.

Functions and Their Uses

1. add(String item)

Use:

- Inserts elements into the custom collection.
- Builds the collection before iteration.

2. iterator()

Use:

- Returns a new iterator object.
- Starts the traversal from the first element.

3. hasNext()

Use:

- Checks whether another element exists.
- Returns true if elements remain.
- Prevents errors caused by reading beyond the collection.

4. next()

Use:

- Returns the current element.
- Automatically moves to the next element.
- Throws NoSuchElementException if no elements remain.

5. remove()

Use:

- Normally removes the current element from a collection.
- In this program, it throws UnsupportedOperationException because the iterator is designed to be read-only.
- Prevents accidental deletion of data.

6. currentIndex

Use:

- Keeps track of the iterator's current position.
- Ensures elements are returned in the correct order.

Real-World Use

Custom iterators are commonly used in:

- Music playlists (play next song)
- Image galleries (view next image)
- Student record systems
- Inventory management
- File explorers
- Online shopping product lists